

Chunk-Length Switching Outside RoboCasa: LIBERO and MimicGen Results

Long-horizon adaptive chunking project

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Setup

Same recipe as the RoboCasa study, on diffusion policies. We label demonstration states with the open-loop stability exponent λ (perturb the first action, replay the chunk open loop, fit the log divergence slope), train small predictor heads on the labels (vision, proprioception, and V-JEPA feature variants), and evaluate fixed chunk lengths $k \in \{16, 8, 4, 1\}$ against switching ($k_{\text{stable}}, k_{\text{unstable}}$) with (16, 4) and (16, 1), plus (8, $\cdot$) completions on the tasks where fixed 8 is competitive. Heads are queried at every replan. 50 episodes per cell.

LIBERO

Task	Fixed k				vjepa			vision			proprio		
	16	8	4	1	(16,4)	(16,1)	(8,1)	(16,4)	(16,1)	(8,1)	(16,4)	(16,1)	(8,1)
Kitchen 3 moka pot	.72	.44	.22	.08	.36	.40	.20	.56	.42	.24	.30	.10	.10
Kitchen 4 bowl in drawer	.76	.84	.76	.46	.54	.62	.82	.64	.58	.72	.58	.38	.58
Kitchen 6 mug in microwave	.00	.00	.00	.00	.00	.00	–	.00	.00	–	.00	–	–
Living Room 2 soup and sauce	.00	.00	.00	.00	.00	.00	–	.00	.00	–	.00	.00	–
Living Room 2 cheese and butter	.00	.00	.00	.00	.00	.00	–	.00	.00	–	.00	.00	–

Table 1: LIBERO success rates. Bold is the best cell per task. Kitchen 6 is zero at every completed cell (its proprio (16, 1) cell is still running); dashes were not run.

MimicGen

Task	Fixed k				full				vision		proprio	
	16	8	4	1	(16,4)	(16,1)	(8,4)	(8,1)	(16,4)	(16,1)	(16,4)	(16,1)
Nut assembly	.34	.42	.42	.52	.40	.32	.34	.42	.34	.46	.44	.38
Coffee preparation	.18	.24	.10	.04	.16	.10	.24	.06	.12	.12	.02	.10

Table 2: MimicGen success rates with in-domain heads. Bold is the best cell per task.

Reading

The fixed- k profiles match the label story. Kitchen 3 collapses from .72 to .08 as the chunk shrinks, the signature of a task where frequent replanning itself destabilizes the policy. Nut assembly is the opposite: mostly unstable, so the shortest chunk is best at .52. Coffee preparation needs a middle chunk (.24 at $k=8$) and dies at $k=1$.

Learned switching never beats the best fixed chunk on these platforms, though from the right base it can now match it. The $(8, \cdot)$ completions sharpen the picture: Kitchen 4’s $vjepa(8, 1)$ reaches .82 against the best fixed .84, and coffee’s full-head $(8, 4)$ reaches .24, exactly its best fixed chunk. Nut assembly’s $(8, 1)$ lands on fixed 8 at .42, still below fixed 1 at .52. The closed-loop unstable tasks pay for every false positive: all three Kitchen 3 $(8, 1)$ cells (.10 to .24) fall far below fixed 16 at .72, and coffee $(8, 1)$ collapses to .06. The pattern is the same one we measured directly on RoboCasa: the heads rank states well offline (AUROC .78 to .89) but the stability signal shifts between demonstration states and rollout states, so the threshold fires in the wrong places. On a task like Kitchen 3 the cost is largest, because every false positive buys a step of the closed-loop instability that made short chunks fail there in the first place; its $proprio(16, 1)$ cell lands at .10, essentially the fixed $k=1$ result.

In the two-axis taxonomy: Kitchen 3 and coffee sit in the closed-loop unstable row where false positives are expensive and long or middle chunks are mandatory. Nut assembly is mostly open-loop unstable, and the right answer is a short fixed chunk rather than switching. Kitchen 6 and both Living Room 2 tasks are at zero everywhere, unstable on both axes, and no chunk length can help. These results complete the negative half of the paper’s claim: the taxonomy predicts not only where switching wins on RoboCasa but also where it must lose elsewhere.